

TPS54360-Based 12 V $\rightarrow$ 5 V, 3 A Buck Converter

Complete Design Calculations

Electrical Engineering Portfolio Project
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Design Specifications

Parameter	Value
Input voltage V_{in}	10.8 V – 13.2 V (12 V nominal)
Output voltage V_{out}	5 V $\pm 2\%$
Output current I_{out}	0 A – 3 A
Output ripple	< 50 mV pp
Transient deviation	± 200 mV (4% of 5 V), load step 0.5 A $\rightarrow$ 3 A
Switching frequency f_{sw}	500 kHz
Topology	Asynchronous buck (HS FET + external Schottky)
Target efficiency	$> 90\%$ at full load

Key TPS54360 parameters (datasheet SLVSBB4G): $V_{\text{REF}} = 0.8$ V, $R_{\text{DS(on)}} = 92$ m Ω , $t_{\text{on,min}} = 135$ ns, current limit ≥ 4.5 A, $g_{m,\text{ea}} = 350$ $\mu\text{A V}^{-1}$, $g_{m,\text{ps}} = 12$ A V^{-1} .

1 Operating Point

Ideal duty cycle

$$D_{\text{ideal}} = \frac{V_{\text{out}}}{V_{\text{in}}} \quad (1)$$

V_{in}	D_{ideal}
10.8 V	0.4630
12.0 V	0.4167
13.2 V	0.3788

Real duty cycle (asynchronous buck with Schottky)

$$D = \frac{V_{\text{out}} + V_F + I_{\text{out}} R_{dc}}{V_{\text{in}} - I_{\text{out}} R_{\text{DS(on)}} + V_F} \quad (2)$$

At $V_{\text{in}} = 12$ V, $I_{\text{out}} = 3$ A, $R_{\text{DS(on)}} = 92$ m Ω , $V_F = 0.5$ V, $R_{dc} \approx 0$:

$$D = \frac{5 + 0.5 + 0}{12 - 3(0.092) + 0.5} = \frac{5.5}{12.224} = 0.450 \quad (3)$$

Real D ranges from 0.499 at $V_{\text{in}} = 10.8 \text{ V}$ to 0.410 at $V_{\text{in}} = 13.2 \text{ V}$. The value $D = 0.450$ is used for downstream loss calculations.

Switching period and timing

$$T_{\text{sw}} = \frac{1}{f_{\text{sw}}} = \frac{1}{500 \text{ kHz}} = 2.0 \mu\text{s} \quad (4)$$

$$t_{\text{on}} = D \cdot T_{\text{sw}} = 0.450 \times 2.0 \mu\text{s} = 900 \text{ ns} \quad (5)$$

$$t_{\text{off}} = (1 - D)T_{\text{sw}} = 0.550 \times 2.0 \mu\text{s} = 1100 \text{ ns} \quad (6)$$

Minimum on-time check

Worst case at $V_{\text{in}} = 13.2 \text{ V}$ ($D = 0.410$):

$$t_{\text{on,min}} = 0.410 \times 2.0 \mu\text{s} = 820 \text{ ns}, \quad \text{margin} = \frac{820}{135} = 6.1 \times \checkmark \quad (7)$$

2 Inductor Selection

Minimum inductance (datasheet Eq. 28, $K_{\text{IND}} = 0.3$)

$$L_{\text{min}} = \frac{V_{\text{out}}(V_{\text{in,max}} - V_{\text{out}})}{I_{\text{out}} \cdot K_{\text{IND}} \cdot V_{\text{in,max}} \cdot f_{\text{sw}}} = \frac{5(13.2 - 5)}{3(0.3)(13.2)(500,000)} = \frac{41}{5.94 \times 10^6} = 6.90 \mu\text{H} \quad (8)$$

Nearest standard value above 6.9 μH : 8.2 μH .

Actual ripple with $L = 8.2 \mu\text{H}$

$$\Delta i_L = \frac{V_{\text{out}}(V_{\text{in,max}} - V_{\text{out}})}{V_{\text{in,max}} \cdot L \cdot f_{\text{sw}}} = \frac{5(8.2)}{13.2(8.2 \times 10^{-6})(500,000)} = \frac{41}{54.12} = 0.757 \text{ A} \quad (9)$$

This is $0.757/3 = 25.2\%$ of I_{out} , below the 30% target. $\checkmark$

Peak and RMS inductor current

$$I_{L,\text{peak}} = I_{\text{out}} + \frac{\Delta i_L}{2} = 3 + \frac{0.757}{2} = 3.38 \text{ A} \quad (10)$$

$$I_{L,\text{RMS}} = \sqrt{I_{\text{out}}^2 + \frac{(\Delta i_L)^2}{12}} = \sqrt{9 + \frac{0.757^2}{12}} = \sqrt{9.048} = 3.01 \text{ A} \quad (11)$$

Current-limit margin: $4.5/3.38 = 1.33 \times$. $\checkmark$

Selected part

Coilcraft XAL7030-822MEC: 8.2 μH , $I_{\text{sat}} = 8.9 \text{ A}$, $I_{\text{rms}} = 6.4 \text{ A}$, $\text{DCR}_{\text{max}} = 19.1 \text{ m}\Omega$. DCR loss: $P_{\text{DCR}} = I_{L,\text{RMS}}^2 \cdot \text{DCR} = 3.01^2(0.0191) = 0.173 \text{ W}$.

3 Output Capacitor

Criterion 1 — loading transient (Eq. 32)

$$C_{\text{OUT},1} \geq \frac{2 \Delta I_{\text{OUT}}}{f_{\text{sw}} \cdot \Delta V_{\text{out}}} = \frac{2(2.5)}{500,000(0.2)} = 50 \mu\text{F} \quad (12)$$

Criterion 2 — unloading transient (Eq. 33)

$$C_{\text{OUT},2} \geq L \cdot \frac{I_{OH}^2 - I_{OL}^2}{V_f^2 - V_i^2} = 8.2 \times 10^{-6} \cdot \frac{3^2 - 0.5^2}{5.2^2 - 5^2} = 8.2 \times 10^{-6} \cdot \frac{8.75}{2.04} = 35.2 \mu\text{F} \quad (13)$$

Criterion 3 — steady-state ripple (Eq. 34)

$$C_{\text{OUT},3} \geq \frac{1}{8f_{\text{sw}}} \cdot \frac{\Delta i_L}{V_{\text{ripple}}} = \frac{1}{8(500,000)} \cdot \frac{0.757}{0.050} = 3.78 \mu\text{F} \quad (14)$$

ESR limit (Eq. 35) and RMS current (Eq. 36)

$$R_{\text{ESR}} \leq \frac{V_{\text{ripple}}}{\Delta i_L} = \frac{0.050}{0.757} = 66 \text{ m}\Omega \quad I_{\text{COUT,RMS}} = \frac{\Delta i_L}{\sqrt{12}} = \frac{0.757}{3.464} = 219 \text{ mA} \quad (15)$$

Dominant criterion: loading transient $\Rightarrow$ 50 μF effective. With $\sim 30\%$ ceramic DC-bias derating, nameplate $\geq 50/0.7 = 71 \mu\text{F}$. **Selected:** $2 \times 47 \mu\text{F}$, 16 V, **X5R, 1210** (Murata GRM32ER61C476ME15L), giving $\sim 66 \mu\text{F}$ effective.

4 Input Capacitor

RMS ripple current (Eq. 38)

Worst case at $V_{\text{in}} = 10.8 \text{ V}$ ($D = 0.5$):

$$I_{\text{CIN,RMS}} = I_{\text{out}} \sqrt{\frac{V_{\text{out}}}{V_{\text{in}}} \cdot \frac{V_{\text{in}} - V_{\text{out}}}{V_{\text{in}}}} = 3 \sqrt{0.463 \times 0.537} = 3 \sqrt{0.2486} = 1.50 \text{ A} \quad (16)$$

Required capacitance (Eq. 39, $\Delta V_{\text{in}} = 250 \text{ mV}$)

$$C_{\text{IN}} \geq \frac{I_{\text{out}}(0.25)}{f_{\text{sw}} \cdot \Delta V_{\text{in}}} = \frac{3(0.25)}{500,000(0.25)} = 6 \mu\text{F} \quad (17)$$

With derating, nameplate $\geq 6/0.75 = 8 \mu\text{F}$. **Selected:** $2 \times 4.7 \mu\text{F}$, 25 V, **X7R, 1210** (Murata GRM32ER71E475KA88L), 9.4 μF nameplate, $\sim 7 \mu\text{F}$ effective, 750 mA RMS per cap.

5 Bootstrap Capacitor

Per datasheet: $C_{\text{BOOT}} = 100 \text{ nF}$, X7R, $\geq 10 \text{ V}$ rated. Selected: Murata GCM188R71E104KA37D (100 nF, 25 V, 0603).

6 Feedback Divider

With $R_{\text{LS}} = 10 \text{ k}\Omega$ (datasheet Eq. 3):

$$R_{\text{HS}} = R_{\text{LS}} \cdot \frac{V_{\text{out}} - 0.8}{0.8} = 10 \text{ k}\Omega \cdot \frac{4.2}{0.8} = 52.5 \text{ k}\Omega \rightarrow 52.3 \text{ k}\Omega \text{ (E96)} \quad (18)$$

Actual output: $V_{\text{out}} = 0.8 \left(1 + \frac{52.3}{10}\right) = 4.984 \text{ V}$ (0.3% error). $\checkmark$

7 Frequency-Set Resistor

$$R_T = \frac{101,756}{f_{\text{sw}}^{1.008}} = \frac{101,756}{500^{1.008}} = 193.7 \text{ k}\Omega \rightarrow 191 \text{ k}\Omega \text{ (E96)} \quad (19)$$

Verification: $f_{\text{sw}} = 92,417/191^{0.991} = 502 \text{ kHz}$ (0.4% high). $\checkmark$

8 UVLO Divider

Targets: $V_{\text{START}} = 9.0 \text{ V}$, $V_{\text{STOP}} = 8.0 \text{ V}$. With $I_{\text{HYS}} = 3.4 \mu\text{A}$, $I_1 = 1.2 \mu\text{A}$, $V_{\text{ENA}} = 1.2 \text{ V}$:

$$R_{\text{UVLO1}} = \frac{V_{\text{START}} - V_{\text{STOP}}}{I_{\text{HYS}}} = \frac{1.0}{3.4 \times 10^{-6}} = 294 \text{ k}\Omega \quad (20)$$

$$R_{\text{UVLO2}} = \frac{V_{\text{ENA}}}{\frac{V_{\text{START}} - V_{\text{ENA}}}{R_{\text{UVLO1}}} + I_1} = \frac{1.2}{\frac{7.8}{294,000} + 1.2 \times 10^{-6}} = \frac{1.2}{27.7 \times 10^{-6}} = 43.3 \text{ k}\Omega \quad (21)$$

→ E96 values: $R_{\text{UVLO1}} = 294 \text{ k}\Omega$, $R_{\text{UVLO2}} = 43.2 \text{ k}\Omega$.

9 Schottky Catch Diode

Selected: B560C-13-F (60 V, 5 A, $V_F = 0.55 \text{ V}$ at 3 A).

$$P_D = I_{\text{out}} \cdot V_F \cdot (1 - D) = 3(0.55)(0.55) = 0.91 \text{ W} \quad (22)$$

10 Plant Transfer Function

Form (datasheet §7.3.15):

$$\frac{V_{\text{out}}}{V_C}(s) = A_{dc} \cdot \frac{1 + s/(2\pi f_Z)}{1 + s/(2\pi f_P)} \quad (23)$$

DC gain

$$R_L = \frac{V_{\text{out}}}{I_{\text{out}}} = \frac{5}{3} = 1.667 \Omega, \quad A_{dc} = g_{m,\text{ps}} \cdot R_L = 12 \times 1.667 = 20.0 \text{ V V}^{-1} \text{ (26 dB)} \quad (24)$$

Dominant pole

$$f_P = \frac{1}{2\pi R_L C_{\text{out}}} = \frac{1}{2\pi(1.667)(66 \times 10^{-6})} = 1447 \text{ Hz} \quad (25)$$

ESR zero

$$f_Z = \frac{1}{2\pi R_{\text{ESR}} C_{\text{out}}} = \frac{1}{2\pi(1.5 \times 10^{-3})(66 \times 10^{-6})} = 1.61 \text{ MHz} \quad (26)$$

At the target crossover $f_{co} = 50 \text{ kHz}$: plant gain = $A_{dc}(f_P/f_{co}) = 20(1447/50000) = 0.579 \text{ V V}^{-1}$ (−4.74 dB), phase $\approx -90^\circ$.

11 Type II Compensator

R_{COMP} — crossover gain (Eq. 48)

$$R_{\text{COMP}} = \left(\frac{2\pi f_{co} C_{\text{out}}}{g_{m,\text{ps}}} \right) \left(\frac{V_{\text{out}}}{V_{\text{REF}} g_{m,\text{ea}}} \right) = \left(\frac{2\pi(50,000)(66 \times 10^{-6})}{12} \right) \left(\frac{5}{0.8(350 \times 10^{-6})} \right) \quad (27)$$

$$= (1.728)(17,857) = 30\,858 \Omega \rightarrow 30.9 \text{ k}\Omega \text{ (E96)} \quad (28)$$

C_{COMP} — zero at plant pole (Eq. 49)

$$C_{\text{COMP}} = \frac{1}{2\pi R_{\text{COMP}} f_P} = \frac{1}{2\pi(30,900)(1447)} = 3.56 \text{ nF} \rightarrow 3.3 \text{ nF} \quad (29)$$

Resulting zero: $1/(2\pi(30,900)(3.3 \times 10^{-9})) = 1560 \text{ Hz}$ (cancels plant pole at 1447 Hz).

C_{HF} — high-frequency pole (Eq. 50/51, take larger)

$$C_{\text{HF,A}} = \frac{C_{\text{out}} R_{\text{ESR}}}{R_{\text{COMP}}} = \frac{(66 \times 10^{-6})(1.5 \times 10^{-3})}{30,900} = 3.2 \text{ pF} \quad (30)$$

$$C_{\text{HF,B}} = \frac{1}{\pi R_{\text{COMP}} f_{\text{sw}}} = \frac{1}{\pi(30,900)(500,000)} = 20.6 \text{ pF} \quad (31)$$

$\rightarrow C_{\text{HF}} = 22 \text{ pF}$ (C0G). Resulting pole: $1/(2\pi(30,900)(22 \times 10^{-12})) = 234 \text{ kHz}$.

Verified loop performance

Crossover $\approx 47 \text{ kHz}$, phase margin $\approx 62^\circ$, gain margin large (Type II does not reach -180° in band).

12 Loss Budget and Efficiency

Loss source	Power (W)	% of total
Schottky conduction	0.908	51
HS FET conduction	0.375	21
HS FET switching	0.180	10
Inductor DCR	0.173	10
Inductor core (est.)	0.050	3
Gate drive	0.018	1
Controller quiescent	0.018	1
Capacitor ESR	0.005	<1
Total	1.73	100

$$\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} = \frac{15}{15 + 1.73} = 89.7\% \quad (32)$$